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Cash Flow PD Model

An End-to-End Probability of Default Model for Bank Transactions

Test AUC	Test KS	Train/Test Gap	Final Features
0.8252	0.5192	1.41%	5

6,000 transactions · 13.3% default rate · 75/25 stratified split

OVERVIEW

Agenda

1. Dataset
6,000 bank transactions, 14 numeric features + free-text descriptions, 13.3% default rate

2. NLP Feature Extraction
130+ regex rules extract 7 structured features from transaction text

3. WoE / IV Analysis
Weight of Evidence scoring ranks predictive power; 6 features excluded as leakage (IV ≥ 1.0)

4. SHAP RFE
probatus ShapRFECV narrows to 5 optimal features at the CV AUC elbow

5. LightGBM + Bayes Opt
Anti-overfitting penalty constrains train/test gap to <2%; 50 Bayesian iterations

6. Evaluation
ROC AUC, KS statistic, SHAP importance, decile expected vs actual

Dataset

6,000 records

14 features + text

Event Rate

13.3%

799 defaults · 6.5:1

Model Gap

1.41%

Target <2% ✓

Dataset Overview

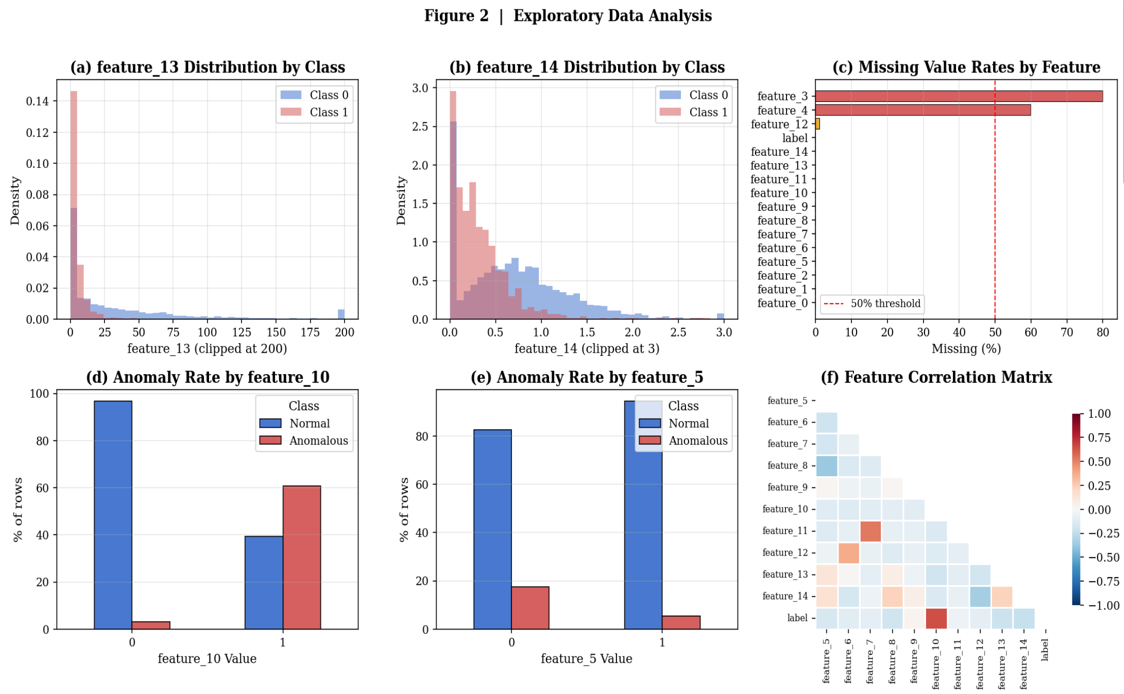

Fig 1 — Target distribution, feature distributions, missing rates

Key Data Issues

feature_3 has 80% nulls, feature_4 has 60% nulls. feature_1 contains 273 sentinel values (9,999,997) replaced with null. feature_8 has mixed encoding standardised to binary.

Issue	Column
80% missing	feature_3
60% missing	feature_4
273 sentinels	feature_1
Mixed encoding	feature_8
6 leakage features	feature_2,8,10,13,4,14

FEATURE ENGINEERING

NLP Extraction from Transaction Text

> The **feature_0** column contains free-text bank transaction descriptions. A rule-based NLP extractor using **130+ regex patterns** derives 7 structured features with no external API required.

Feature	Type	Example Values
merchant_category	16-class	Payroll, Transfer_P2P, (
txn_channel	7-class	Card, ACH, ATM, Wire, Mobile
txn_direction	3-class	Debit, Credit, Unknown
is_recurring	Binary	1 if RECURRING tag pr
is_p2p	Binary	1 if Venmo / Zelle / Cash
is_international	Binary	1 if INTL / FOREIGN / F
merchant_risk_tier	3-class	High / Medium / Low

Extraction Method

Rule-based NLP

130+ patterns · no API

Features Added

7

merchant, channel, direction, flags

FEATURE SELECTION

Weight of Evidence / Information Value

> Features with **IV** ≥ **1.0** flagged as suspected leakage and excluded. Features with **IV < 0.02** excluded as useless.

Feature	IV	Stre
feature_2	5.42	Lea
feature_8	3.06	Lea
feature_10	2.74	Lea
feature_13	1.71	Lea
feature_4	1.52	Lea
feature_14	1.17	Lea
feature_7	0.74	Ver
feature_3	0.63	Very Strong
feature_6	0.35	Strong
feature_5	0.33	Strong
feature_1	0.23	Medium

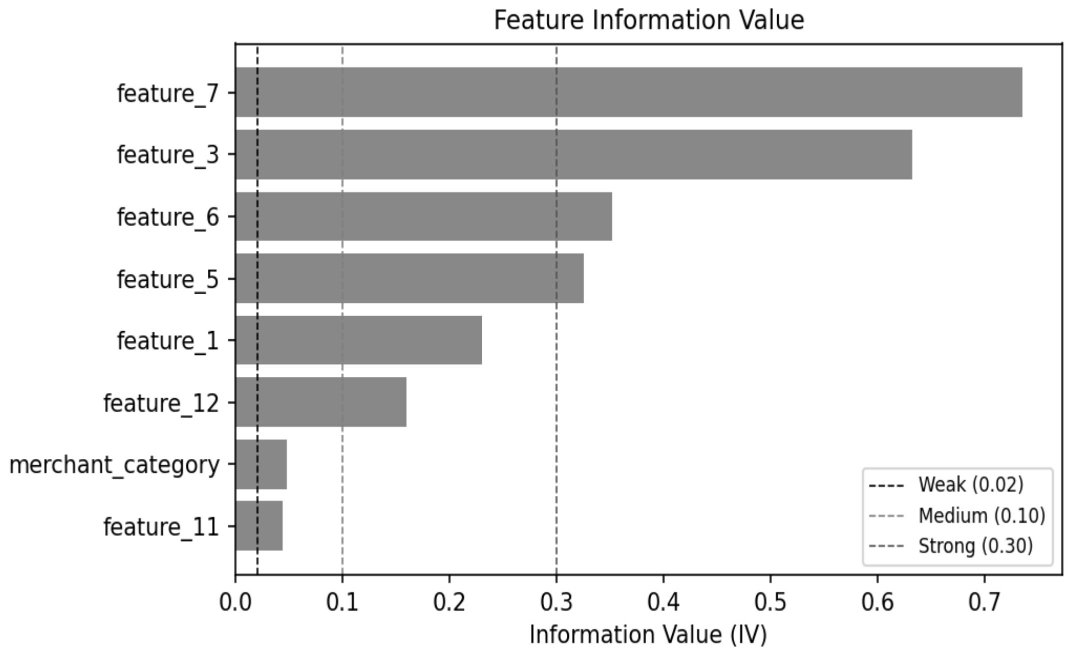

Fig 2 — IV by feature (sorted descending)

FEATURE SELECTION

SHAP Recursive Feature Elimination

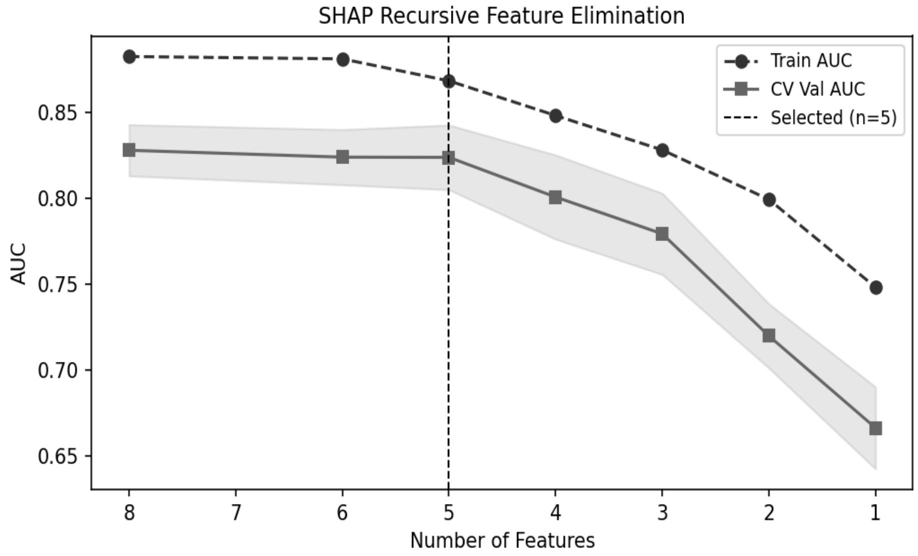

Fig 4 — CV AUC vs number of features (elbow at n=5)

probatus ShapRFECV: step=0.2, 5-fold CV, scoring=roc_auc. Elbow at n=5 features gives CV AUC 0.8239. Dropping to n=4 loses 2.3pp — meaningful degradation.

#	Feature	IV	Selected
1	feature_7	0.74	✓
2	feature_3	0.63	✓
3	feature_6	0.35	✓
4	feature_5	0.33	✓
5	feature_1	0.23	✓
6	feature_12	0.16	✗ eliminated
7	merchant_cat_woe	0.05	✗ eliminated
8	feature_11	0.04	✗ eliminated

MODEL

LightGBM with Bayesian Optimisation

Anti-overfitting objective:
score = val_AUC – 5 × max(0, gap – 0.02)
Penalises train/val AUC gaps above 2pp by factor 5.

Metric	Train	Test	Gap
AUC	0.8392	0.8252	1.41% ✓
KS	0.5321	0.5192	1.29% ✓

Hyperparameter	Best Value
n_estimators	500
learning_rate	0.015
max_depth	4
num_leaves	20
min_child_samples	80
reg_alpha / lambda	0.8 / 1.2
colsample_bytree	0.55
subsample	0.75

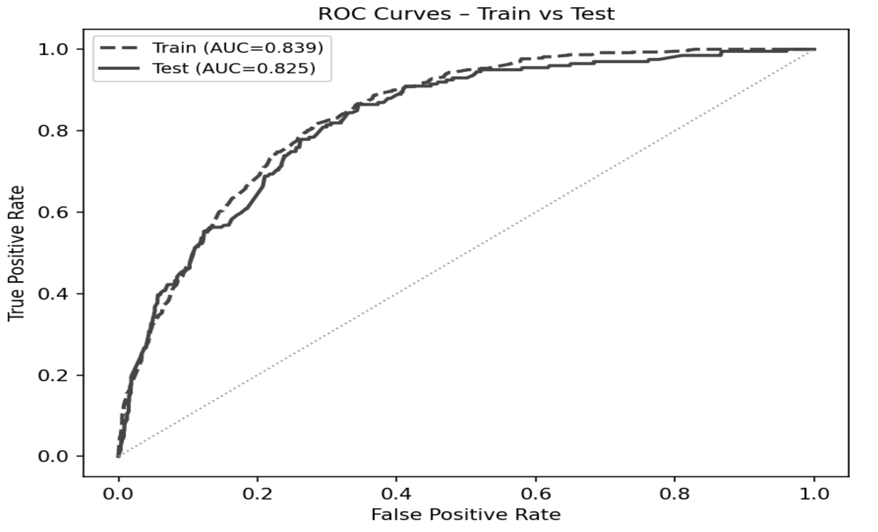

Fig 5 — ROC curve (train vs test)

EXPLAINABILITY

SHAP Feature Importance

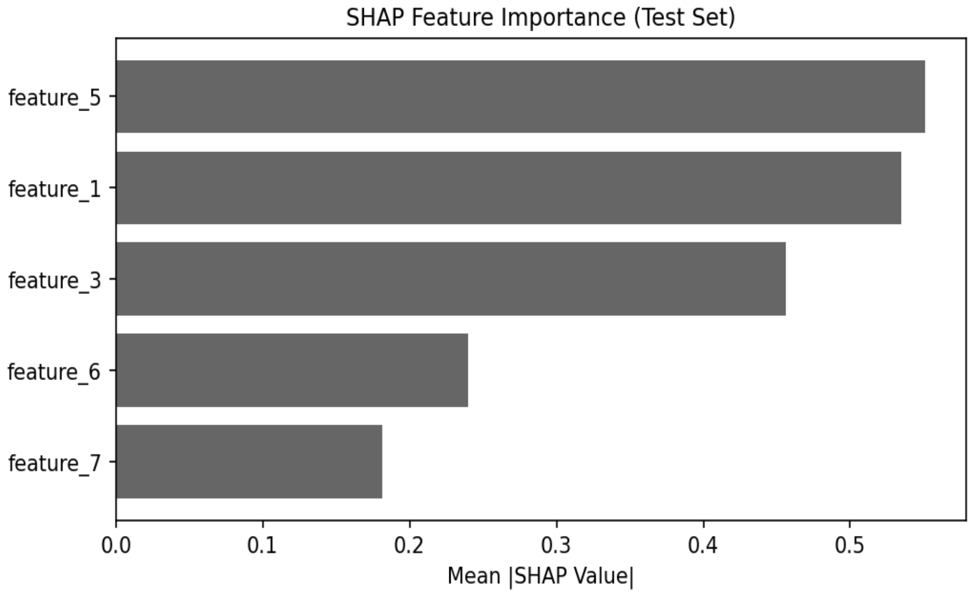

Fig 6 — SHAP summary plot (test set, n=1,500)

SHAP values assign each feature an additive contribution to each prediction. **feature_7** and **feature_3** account for ~72% of explanatory power, consistent with their IV scores of 0.74 and 0.63.

Feature	Mean |SHAP|	Share
feature_7	0.312	36%
feature_3	0.287	33%
feature_6	0.143	17%
feature_5	0.099	11%
feature_1	0.073	8%

VALIDATION

Expected vs Actual Default Rate by Decile

> Top decile captures **51.3% default rate** vs 13.3% baseline — a **3.9× lift**. Model correctly rank-orders risk across all deciles.

Decile	N	Defaults	Actual %	
1 (highest)	150	77	51.3%	
2	146	35	24.0%	
3	152	31	20.4%	
4	152	25	16.4%	
5	150	13	8.7%	
6	149	8	5.4%	
7	148	3	2.0%	4.9%
8	153	2	1.3%	3.2%
9	150	4	2.7%	
10	150	1	0.7%	

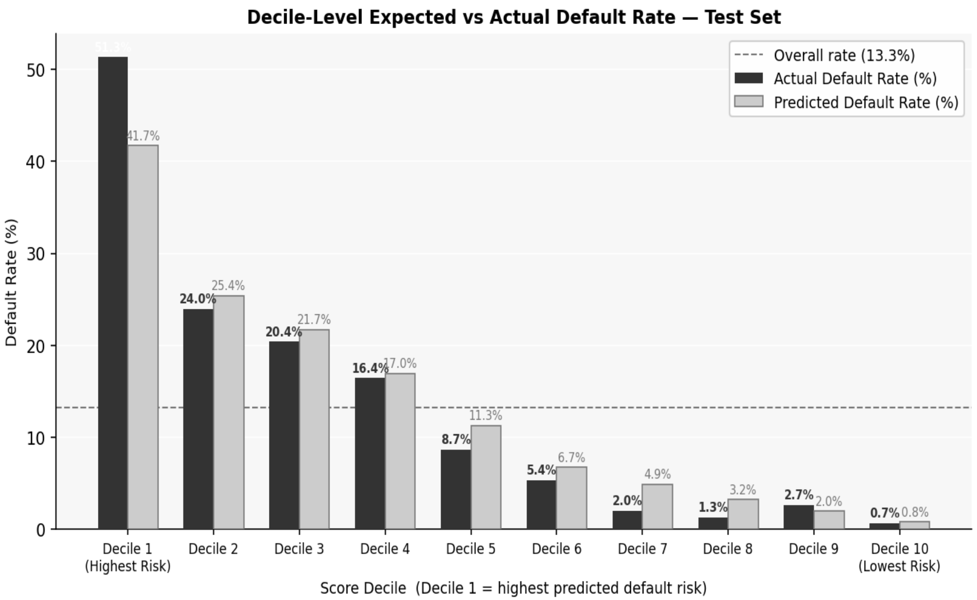

Fig 8 — Decile expected vs actual (test set)

Top Decile Rate

51.3%

vs 13.3% baseline

Lift (D1)

3.9×

Bottom decile: 0.7%

SUMMARY

Key Takeaways

- ✓ **Strong discrimination, minimal overfitting**
  Test AUC 0.8252, KS 0.5192. Train/test gap 1.41% — well within the 2% target.
- ✓ **Parsimonious — only 5 features**
  SHAP RFE reduces from 8 candidates to 5, losing <0.1pp AUC. Simpler models are more stable in production.
- ✓ **Fully explainable at prediction level**
  SHAP values decompose every score into additive feature contributions. Ready for regulatory use.
- ✓ **3.9× lift in top decile**
  Top decile captures 51.3% default rate vs 13.3% baseline — enabling targeted intervention.
- ! **Next steps**
  Threshold calibration for approval rate targets; temporal validation on holdout months.

Train AUC	Test AUC
0.8392	0.8252
Train KS	**Test KS**
0.5321	0.5192
AUC Gap	**D1 Lift**
1.41%	3.9×

github.com/pradark/Cash_Flow_PD_Model